

Civil Engineering

Apply Algebra and Geometry

Assignment 1 - Questions Only

Instructions: Answer all questions. Show all working and use correct units.

Section A

1. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$a^m \times a^n = a^{m+n}$$

2. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\log(ab) = \log a + \log b$$

3. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

4. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Section B

5. Solve equations arising from road gradient calculations. **(10 marks)**
6. Calculate distance between two control points. **(10 marks)**
7. Find the slope of a line joining two points. **(10 marks)**
8. Create one realistic civil engineering problem based on this topic and solve it fully. **(20 marks)**